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Measurement of neural respiratory drive in patients with COPD

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Abstract

Assessment of neural respiratory drive may be useful in patients with chronic obstructive pulmonary disease (COPD) for diverse clinical and academic reasons. We hypothesised that the oesophageal diaphragm EMG during CO_2 rebreathing and treadmill exercise could be used for this purpose. The oesophageal catheter consisted of nine consecutive recording electrode coils, which formed five pairs of electrodes with an inter-electrode distance 3.2 cm within a recording pair. Each coil was 1 cm in length and the gap between adjacent coils was 0.5 mm. Maximal isometric contractions at functional residual capacity (FRC) and maximal voluntary inspirations from FRC to total lung capacity (TLC) were performed. All subjects performed CO_2 rebreathing until end-tidal CO_2 was approximately 9% or they became intolerably breathless. There was a good linear relationship between peak of root mean square (RMS) of the diaphragm EMG and end-tidal CO_2 ($r = 0.92 \pm 0.06$) during CO_2 rebreathing. The method was also shown to be feasible during exercise. It is concluded that the diaphragm EMG recorded from an oesophageal electrode is a useful technique to assess neural respiratory drive in patients with COPD.
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Keywords: Diaphragm EMG; CO_2 rebreathing; Neural respiratory drive; COPD

1. Introduction

Neural respiratory drive in chronic obstructive pulmonary disease (COPD) is of interest for both clinical and scientific reasons. For example, neural drive is recognised to be high in patients who subsequently fail to wean from mechanical ventilation (Tobin et al., 1986; Murciano et al., 1988). Available methods for assessing neural drive, including minute ventilation (VE) and mouth occlusion pressure ($P_{0.1}$) (Whitelaw et al., 1975), have limited value in patients with COPD during exercise, particularly if there is dynamic hyperinflation. The diaphragm EMG, recorded from an oesophageal electrode, is an alternative method of assessing neural respiratory drive in normal subjects (Lopata et al., 1977; Luo et al., 2001) and patients with

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COPD (Lourenco and Miranda, 1968; Gorini et al., 1990). However, when using a single pair oesophageal electrode there is concern that electrode movement, resulting from a change in lung volume, may make measurement inaccurate (Daubenspeck et al., 1989). Recently, we (Luo et al., 2000, 2001) and others (Daubenspeck et al., 1989; Beck et al., 1998) have demonstrated that the influence of lung volume on the diaphragm EMG can be greatly reduced by using a multipair oesophageal electrode. However, the multipair oesophageal electrodes used in previous studies usually had a substantial gap between electrode coils, potentially causing some EMG signals to be missed.

Therefore, we hypothesised that a catheter with a smaller gap between multiple paired coils may more reliably record the diaphragm EMG. To test this hypothesis, we constructed a novel multipair oesophageal catheter with 0.5 mm between electrode coils to record the diaphragm EMG and evaluated it for the assessment of central respiratory drive to the diaphragm in patients with COPD during CO_2 rebreathing and assessed its feasibility during exhaustive exercise.

2. Methods

2.1. Subjects

Seven patients with moderately severe COPD participated in the study. All patients were clinically stable and not had an acute exacerbation in the preceding month. No patients had clinically significant coexisting disease. Patients were taking their usual medication, including inhaled bronchodilators. Most patients had an experience of exercising on a treadmill. The study was approved by the local ethics committee and all subjects gave informed consent.

2.2. Oesophageal electrode and its positioning

A multipair oesophageal electrode catheter, modified from a previous design (Luo et al., 2000, 2001) was used to record the diaphragm EMG. The electrode catheter consisted of nine consecutive recording electrode coils (number 1 being proximal and 9 distal); each coil was 10 mm in length and there was a gap of 0.5 mm between adjacent recording electrodes. The external diameter of the catheter was 2.0 mm. Five pairs of electrodes were formed with an inter-electrode distance of 3.2 cm within a recording pair. Pair 1 consisted of coils 1 and 5, pair 2 of coils 2 and 6, pair 3 of coils 3 and 7, pair 4 of coils 4 and 8, and pair 5 of coils 5 and 9. There was another electrode (coil 0), which was connected to ground and was 2 cm away from coil 1 (Fig. 1). The oesophageal electrode was passed through the nose and swallowed into the oesophagus. Coil 5 was positioned closed to the crus of the diaphragm based on the spontaneous EMG recorded from the five pairs of electrodes. The ideal position of the electrode catheter was characterized by large EMG activity from pairs 1 and 5, and a small EMG recorded from electrode pair 3 during resting breathing, as we previously described (Luo et al., 2001). After the oesophageal electrode was satisfactorily positioned, the catheter was securely taped at the nose. The diaphragm EMG signals were amplified and band-pass filtered between 20 Hz and 1 kHz (Magstim Co., Whitland, Dyfed, UK).

2.3. Protocol

After the oesophageal electrode catheter was satisfactorily positioned, the following tests were performed. A minimum of 20 min rest was permitted between each group of tests

2.4. Maximal voluntary EMG

Maximal diaphragm EMG was recorded during two manoeuvres: maximal inspiration from functional residual capacity (FRC) to total lung capacity (TLC) (EMG_{tlc}) and during a maximal static inspiratory effort at FRC against a closed valve (EMG_{static}). Patients were asked to sustain their maximal effort for 1 s. There was at least 30 s between each effort. At least three efforts were made for each manoeuvre, with the subjects seated on a chair. Manoeuvres were performed in random order. The EMG was recorded simultaneously from the five pairs of electrodes.

2.5. CO_2 rebreathing

CO_2 rebreathing was performed using the Read rebreathing method (Read, 1967), described in a

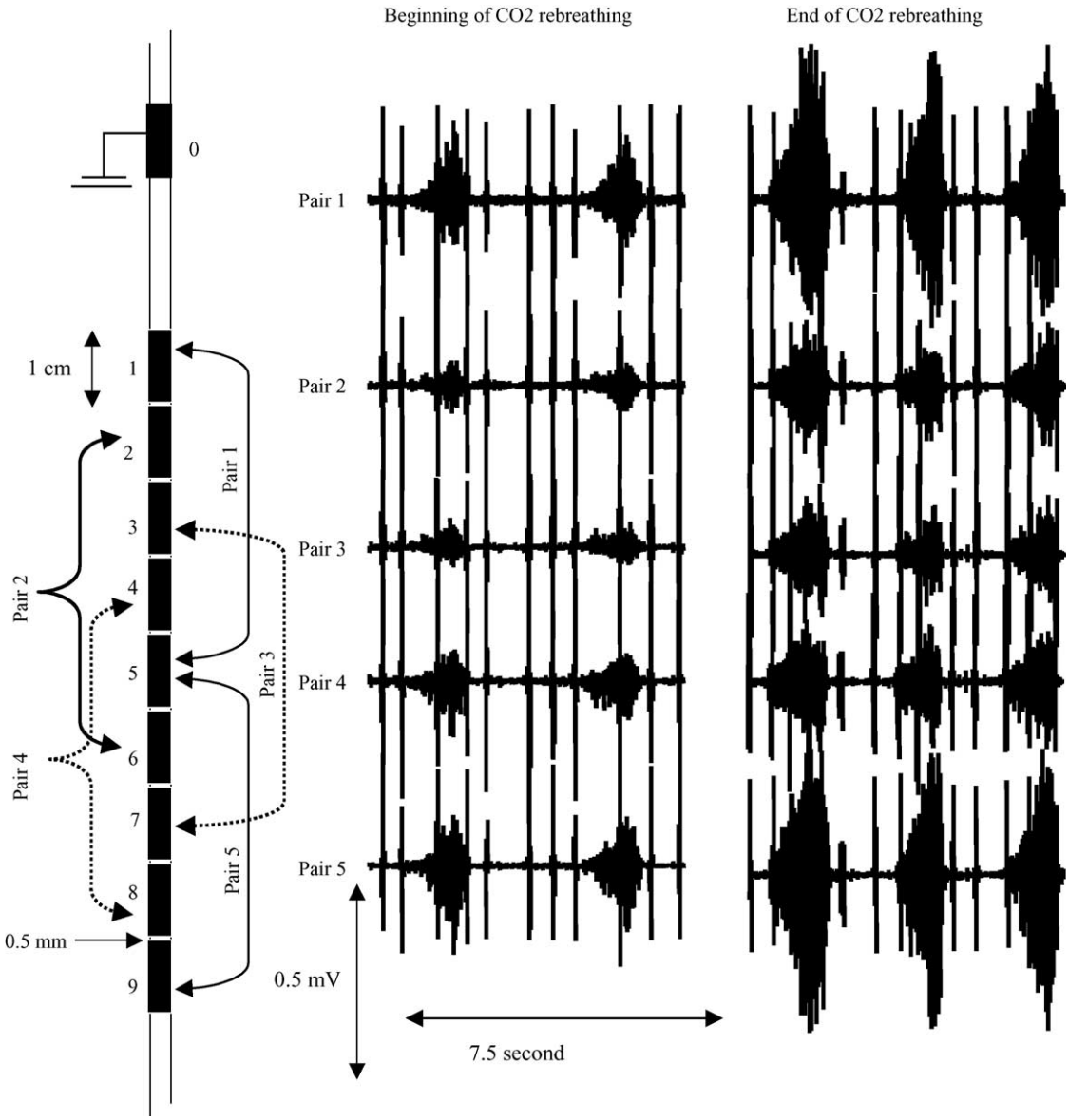

Fig. 1. Oesophageal electrode and the diaphragm EMG recorded from five pairs of electrode during CO_2 rebreathing. The diaphragm EMG increases during CO_2 rebreathing and the EMG at the end of CO_2 rebreathing is much larger than that at the beginning of CO_2 rebreathing. Data from one subject.

previous study (Luo et al., 2001). Subjects were seated on a chair and initially breathed room air through a mouthpiece connected to an open breathing circuit for a few minutes. When subjects were familiar with breathing with a mouthpiece, the rebreathing bag with 5% CO_2 in O_2 was introduced. CO_2 rebreathing was terminated when the end-tidal CO_2 concentration was approximately 9% or subjects could no longer tolerate the test. The diaphragm EMG, end-tidal CO_2 and inspiratory flow were continuously recorded.

Table 1
Anthropometric characteristics and pulmonary function data in patients with COPD

Subject	Age (years)	Sex	Weight (kg)	Height (cm)	FEV1 (% pre)	VC (% pre)	FEV1/VC (%)	RV (% pre)	PO_2 (Kpa)	PCO_2 (Kpa)
1	74	F	60	160	53	96	47	138	8.8	5.5
2	55	M	56	160	29	76	29	210	7.8	6.3
3	78	M	66	162	32	66	36	112	7.4	6.5
4	74	M	70	159	27	79	26	140	8.5	4.7
5	62	M	82	180	29	68	33	192	8.1	6.0
6	79	M	66	168	29	76	27	185	8.4	4.9
7	61	M	74	178	44	98	35	150	9.3	5.1
Mean	69		68	167	35	80	33	161	8.3	5.6
S.D.	9		9	9	10	12	7	35	0.6	0.7

2.6. Treadmill exercise

Subjects breathed through a mouthpiece with a noseclip in place. The speed and gradient for treadmill exercise were chosen to achieve the patient's habitual "brisk" walking speed described in a previous study (Polkey et al., 1995). Inspiratory flow and diaphragm EMG measured from five pairs of electrodes were recorded. Transcutaneous O_2 saturation was monitored with a pulse oximeter during exercise. All subjects performed exercise until unable to continue because of intolerable dyspnea or O_2 saturation less than 80%. Supplemental oxygen (2.0 L/min) was administered when the patient's oxygen saturation prior to exercise was less than 90%.

2.7. Signal recordings

End-tidal CO_2 was monitored by a capnograph (Model 455, P. K. Mogan Ltd., Kent, England). Flow was measured with a pneumotachograph (Fleisch no. 4, Lausanne, Switzerland) connected to a Mercury CS6 electrospirometer (G.M. Engineering, Kilwinning, Scotland, UK). Signals were recorded using PowerLab 8s hardware and software (ADInstruments Pty Ltd., Castle Hill, Australia) combined with a Power Mac computer. The sample rate was 2 kHz for voluntary EMG and 200 Hz for all other signals. Signals were displayed in real time and stored in the computer hard disk for further off-line analysis.

Table 2
Maximal RMS of EMG generated by maximal inspiration and the correlation coefficient (r) of RMS to end-tidal CO_2, minute ventilation (VE) and tidal volume (VT) during CO_2 rebreathing and exercise

Subject	RMS of EMG (μV)		CO_2 rebreathing (r)			Exercise (r)	
	EMG_{static}	EMG_{tlc}	RMS-VT	RMS-VE	RMS-CO_2	RMS-VE	RMS-VT
1	124	135	0.77	0.94	0.96	0.52	0.29
2	179	150	0.93	0.93	0.96	0.24	0.27
3	120	120	0.68	0.88	0.92	0.82	0.53
4	203	212	0.86	0.96	0.94	0.79	0.54
5	173	197	0.95	0.96	0.90	0.52	0.38
6	180	190	0.34	0.77	0.78	0.84	−0.69
7	98	108	0.03	0.88	0.95	0.90	0.64
Mean	154	159	0.65	0.90	0.92	0.66	0.28
S.D.	39	41	0.34	0.07	0.06	0.24	0.45

EMG_{static}, maximal EMG produced by isometric constraction at FRC; EMG_{tlc}, maximal EMG generated by maximal inspiration from FRC to TLC.

2.8. Data analysis and presentation

Data was analyzed off-line. The raw signal was converted to root mean square (RMS) using Powerlab software (ADInstruments Pty Ltd., Australia) as shown in the previous study (Luo et al., 2001). The RMS of the EMG was calculated by computer with a time constant of 100 ms. During CO_2 rebreathing and exercise, peak of RMS of each breath was measured and the mean of five consecutive breaths was calculated. To avoid the influence of the electrocardiogram on the diaphragm EMG, RMS was measured from the segments between QRS complexes. Minute ventilation, tidal volume (VT) were calculated by integrating flow from five consecutive breaths. In the static and rebreathing tests peak of RMS is presented in microvolts, but for the exercise run peak of RMS is normalised to the single largest value obtained during the run.

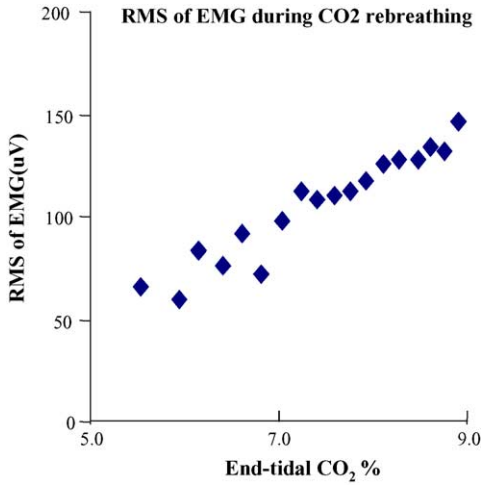

Fig. 2. Peak of RMS increases during CO_2 rebreathing. There is good relationship between end-tidal CO_2 and amplitude of the RMS. No plateau of RMS of EMG is observed. Data from one subject.

3. Results

3.1. Maximal diaphragm EMG measured by different manoeuvres

Anthropometric and lung function data are shown in Table 1. The RMS of the EMG_{tlc} and EMG_{static} are shown in Table 2. The maximal EMG measured from the two manoeuvres was similar: $159 \pm 41\ \mu V$ for EMG_{tlc} and $154 \pm 39\ \mu V$ for EMG_{static}. The electrode pair for recording EMG_{tlc} was usually the same or adjacent to the pair for recording EMG_{static}.

3.2. Diaphragm EMG during CO_2 rebreathing

Ventilation and breathing frequency increased during CO_2 rebreathing. Most patients stopped CO_2 rebreathing when, end-tidal CO_2 reached 9% rather than because of breathlessness. At the end of CO_2 rebreathing, VE was 27.7 ± 7.5 L/min and was significantly different from that (13.0 ± 3.9 L/min) at the beginning of CO_2 rebreathing ($p < 0.01$). However, mean VT at the end of CO_2 rebreathing (1.33 ± 0.40 L) was not statistically larger than that at the beginning (1.02 ± 0.43 L) (p = ns). In some subjects, VT increased little (Subjects

Table 3
Minute ventilation (VE), tidal volume (VT), breathing frequency (f) and heart rate (HR) during exercise and CO_2 rebreathing

Subject	Exercise								CO_2 rebreathing					
	First five breaths				Last five breaths				First five breaths			Last five breaths		
	VT	VE	f	HR	VT	VE	f	HR	VT	VE	f	VT	VE	f
1	0.52	15.4	29	78	0.75	25.1	34	129	0.90	10.52	11	1.00	21.8	22
2	0.87	19.5	22	80	0.95	26.4	28	91	0.82	18.2	22	1.36	30.0	22
3	0.77	17.9	23	92	0.74	20.7	28	105	0.76	11.1	15	0.78	15.4	20
4	0.70	17.4	25	101	0.83	24.2	29	116	0.85	11.8	14	1.19	25.1	21
5	0.75	15.0	20	108	1.59	38.0	24	120	0.56	10.8	19	1.54	35.6	23
6	1.28	22.8	18	108	1.07	35.5	33	135	11	19.0	11	1.46	30.3	21
7	0.85	17.0	20	88	1.61	36.3	23	136	1.55	62.0	6	2.00	36.0	18
Mean	0.82	17.9	23	94	1.08	29.5	28	119	1.02	13.0	14	1.33	27.7	21
S.D.	0.23	2.6	4	12	0.38	6.9	4	17	0.43	3.9	5	0.40	7.5	2

1 and 3) or even decreased (Subject 6) (Table 3). The peak of RMS recorded from all electrode pairs increased during CO_2 rebreathing. No clear plateau of the peak of RMS was observed during CO_2 rebreathing. The electrode pair for recording the maximal EMG at the beginning of CO_2 rebreathing was the same or adjacent to the pair at the end (Fig. 1). There was a good relationship between peak of RMS and end-tidal CO_2 ($r = 0.92 \pm 0.06$) (Fig. 2). The correlation coefficient of RMS to VE ($r = 0.90 \pm 0.07$) was better than that of RMS to VT ($r = 0.65 \pm 0.34$) (Table 2). The mean peak of RMS at the end of CO_2 rebreathing was 90% of peak of RMS of EMG_{tlc}.

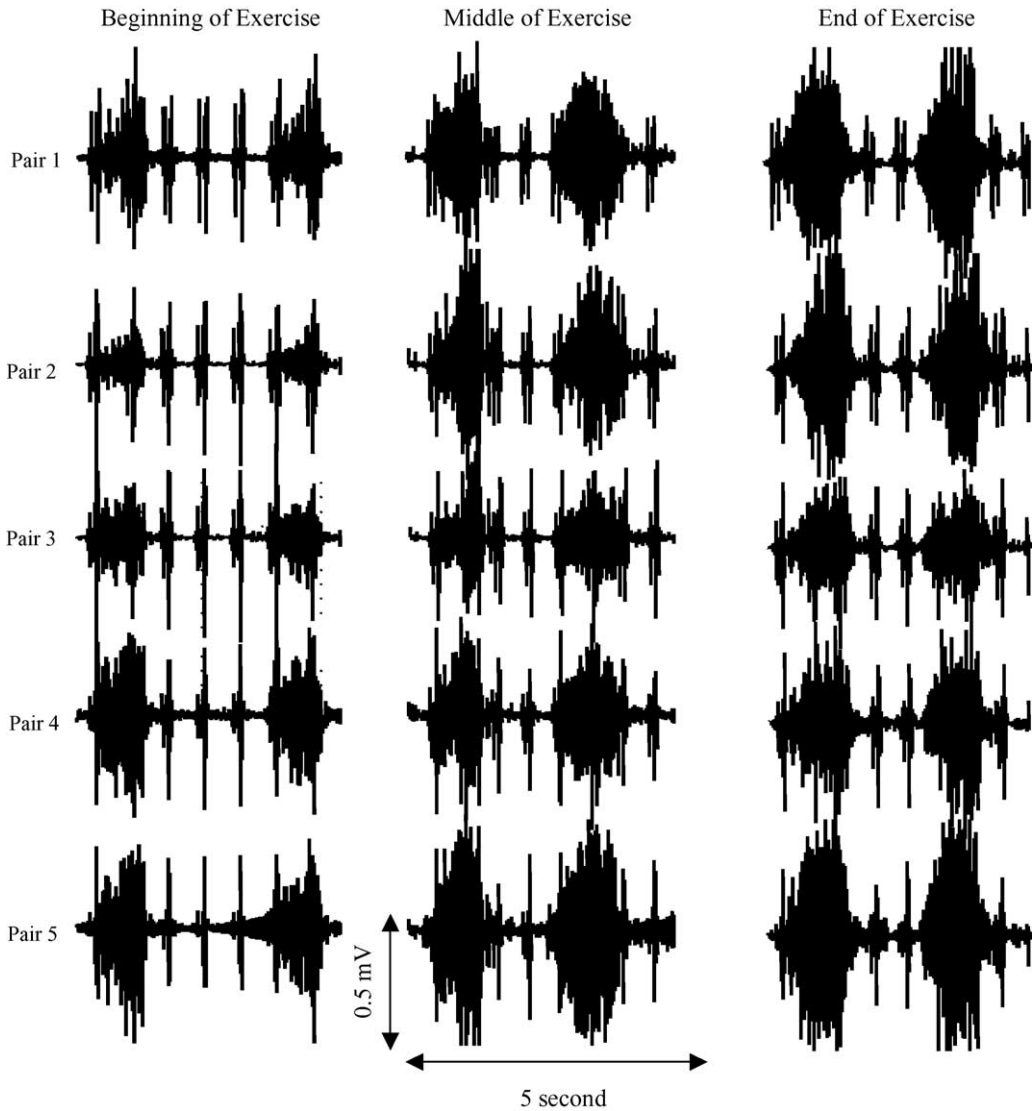

Fig. 3. Diaphragm EMG recorded simultaneously from five pairs of electrode during exercise. The electrode pair for recording maximal EMG was the same over the whole exercise. The EMGs recorded in the middle and at the end of exercise are larger than that at the beginning of exercise. Data from one subject.

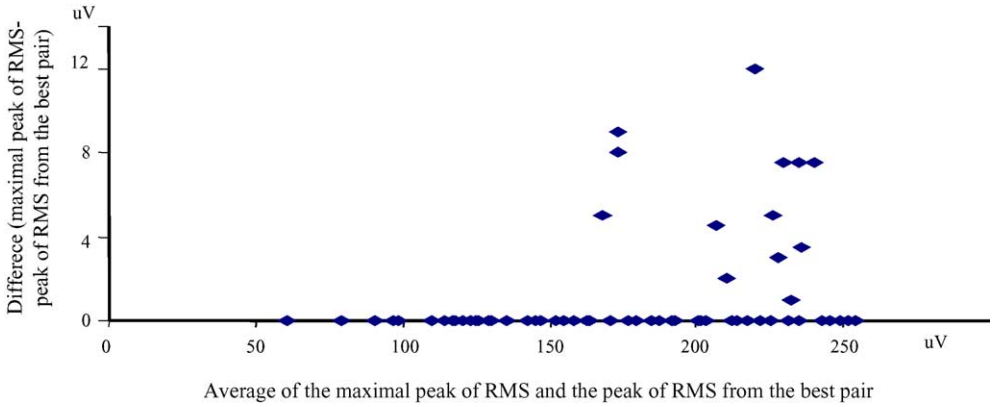

Fig. 4. Bland and Altman (1986). The electrode pair for recording maximal peak of RMS almost remained the same over the study during exercise. The best pair was defined as the pair recording the maximal peak of RMS at the beginning of the study. Data from one subject.

3.3. Neural drive during exercise

All subjects stopped exercise because of breathlessness except for one who was asked to stop because of serious oxygen desaturation. In most subjects, the electrode pair which recorded the maximal EMG at the beginning of exercise also recorded the maximal EMG over the whole exercise (Fig. 3). In some subjects, the electrode pair which recorded the maximal EMG at the end of exercise was adjacent to the electrode pair recording the maximal EMG at the beginning of exercise (Fig. 4). VE increased initially and reached a plateau in all subjects. Peak of RMS recorded from all pairs of electrodes increased with exercise (Fig. 5). There was no significant relationship between RMS and VT during exercise ($r = 0.28 \pm 0.45$).

4. Discussion

The main finding of the present study is that there was a strong linear relationship between the RMS of the EMG and end tidal CO_2 during CO_2 rebreathing in patients with advanced COPD suggesting that the method of data collection and analysis used in this study is an accurate measure of neural drive. Peak of RMS was also increased during another stimulus to ventilation, namely exercise. The fact that the electrode pair for recording maximal EMG remained almost the same during different condition indicates that the crus of the diaphragm only moves slightly, if at all, during CO_2 rebreathing and exercise although dynamic hyperinflation was presumed to have occurred.

4.1. Oesophageal electrode design

Reliable recording of the diaphragm EMG relies on constant configuration of the recording electrode in the diaphragm during spontaneous breathing. Balloon stabilized electrodes do not achieve a constant relationship with the diaphragm (Gandevia and McKenzie, 1986; Beck et al., 1998; Luo et al., 2001). To overcome the influence of change in lung volume on the diaphragm EMG, a multipair oesophageal electrode is required. By selecting the maximal diaphragm EMG on breath by breath basis from five pairs of electrodes, a constant relationship between recording electrodes and the crus of the diaphragm can be achieved. It has been shown that in normal subjects the crus of the diaphragm usually move less than 1.5 cm when lung volume changes from RV to TLC (Luo et al., 2000, 2001). The changes in lung volume in patients with COPD are less than those in normal subjects because of the reduction of inspiratory capacity resulting from an increase in RV. Therefore, the movement of the crus of the diaphragm in patients with COPD will be smaller than in normal subjects. Consistent with this we found that the electrode pair for recording the maximal EMG at the beginning of exercise or CO_2 rebreathing remained the same, or was the adjacent recording pair at the end of exercise or CO_2 rebreathing although dynamic hyperinflation might be presumed to have occurred (O'Donnell

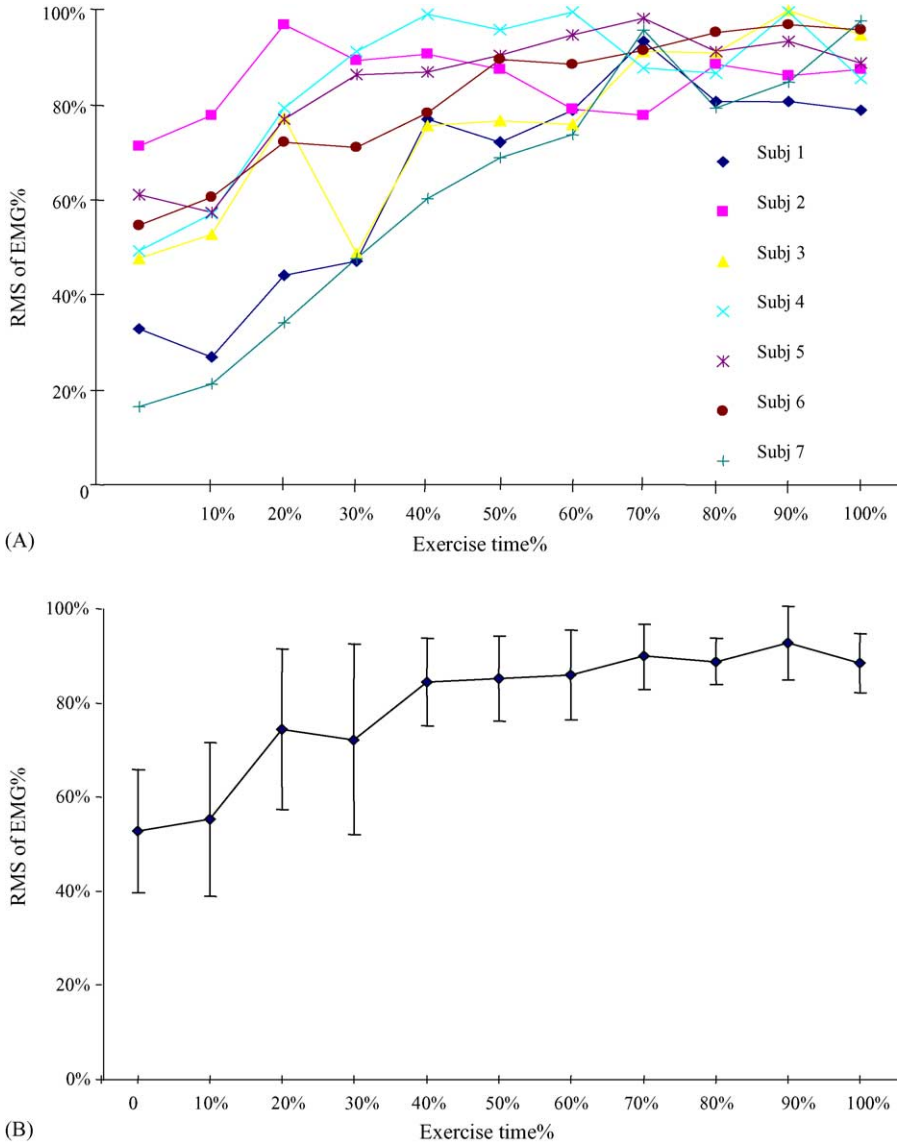

Fig. 5. Peak of RMS normalised with a single maximal breathing during exercise. Diaphragm EMG increases and reaches a plateau during exercise in all subjects: (A) Individual data; (B) mean ± S.D., group data.

et al., 2001). The RMS of the maximal EMG_{static} was similar to that of the maximal EMG_{tlc}, suggesting that change in lung volume has little effect on RMS of the maximal EMG when an electrode can constantly detect the maximal diaphragm EMG. Because the effect of lung volume on the oesophageal diaphragm EMG was believed to be due to change in the distance between electrode and the crus (Gandevia and McKenzie, 1986; Daubenspeck et al., 1989; Beck et al., 1995), it is therefore not surprising that, in the present study, with the multipair electrode whose gap is small, the maximal EMG measured with an isometric contraction at FRC was similar to that measured with a deep inspiration from FRC to TLC.

4.2. Neural respiratory drive assessed by the oesophageal electrode

A number of studies (Lopata et al., 1977; Önal et al., 1979; Luo et al., 2001) have shown that there was a good relationship between diaphragm EMG and end-tidal CO_2 in normal subjects. Recently, we have shown that the diaphragm EMG recorded from a multipair oesophageal electrode could be used to assess neural respiratory drive in normal subjects (Luo et al., 2001). In the present study, a good linear relationship between the RMS of the diaphragm EMG and end tidal CO_2 during CO_2 rebreathing indicates that the technique may be useful in assessing neural drive in patients with COPD. A relatively weak correlation ($r = 0.65$) between the peak of RMS and VT during CO_2 rebreathing particularly during exercise may be due to dynamic hyperinflation, which limits the increase in VT.

4.3. Neural respiratory drive during exercise

Although not the main aim of the present study, examination of Fig. 5 shows that the diaphragm EMG increased initially and reached a plateau after 40% of exercise time. The plateau of the diaphragm EMG could be viewed as indication of submaximal neural drive or alternatively early achievement of maximal drive in patients with COPD. However in Fig. 5, because the values of RMS are normalised with the largest EMG from a single breath during exercise it is not possible to confidently resolve this point.

The finding in the present study differs from that reported by Sinderby et al. (2001) who showed that diaphragm EMG increased gradually, until the end of exercise without a plateau during an incremental cycle test. The difference in changes of EMG and ventilation between two studies may be partly explained by a different type of exercise. It has been recently reported that VO_2 max increased during incremental bicycle exercise without plateau, whereas VO_2 max increased initially and reached plateau during 6 min walk test (Troosters et al., 2002). However, two further factors may also explain differences between my data and those of Sinderby et al. (2001). First, significant differences exist in electrode configuration between the two studies which may also have contributed to the difference in results, particularly if as we hypothesised inter-electrode spacing wields an important influence. Finally, Sinderby et al. (2001) presented their data at just three levels of exercise making detection of a plateau difficult.

No plateau of EMG was observed during CO_2 rebreathing. Some factors may explain these results. Firstly, we set the target of end tidal CO_2 at 9% to avoid severe acidosis. The fact that most subjects stopped CO_2 rebreathing because of the end-tidal CO_2 reached 9% rather than the onset of symptoms suggests that patients might tolerate more intense hypercapnia, but this was precluded by our institutional guidelines. However, an additional consideration is that neural drive activated by pure CO_2 differs from that stimulated by exercise, which also involves hypoxemia and lactic acidosis (Polkey et al., 2000).

In conclusion, there is good relationship between the diaphragm EMG and end-tidal CO_2 during CO_2 rebreathing. Diaphragm EMG recorded from an oesophageal electrode is useful technique in accurately assessing neural respiratory drive. Although we prefer using a multipair oesophageal electrode to constantly record the maximal diaphragm EMG, a single pair oesophageal electrode may also be used to assess change in neural respiratory drive in patients with COPD if it is positioned properly.

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